

Exclusion of the fixed-linear primitive cubic-pencil quartic row

A self-contained, AI-assisted research-program note

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with substantial AI assistance

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Review, priority, and verification notice. This manuscript is not peer reviewed. Exact computer algebra checks the identities and finite case ledgers encoded in the accompanying files; it is evidence, not peer review. Substantial AI assistance was used in the mathematical exploration, proof organization, software, hostile internal audits, and exposition. No claim of worldwide priority is made. The result excludes one frozen quartic leading-form row; it does not prove the full degree-four case of the Jacobian Conjecture.

Abstract

Let $F : \mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ be a degree-four Keller map. We prove that F is an automorphism when the canonical invariant tuple of its quartic homogeneous part is

$$(\text{rank } JH_4, e, a, b, \delta, \nu) = (2, 1, 3, 1, 1, 1).$$

Equivalently, the leading part is an invariant fixed linear divisor times a primitive cubic pencil. A coefficient-independent normalization gives $H_4 = h(p, q, 0)$ and an intrinsic horizontal/unique-vertical split. A homogeneous first-integral valuation settles the horizontal branch and reduces the vertical branch to two cubic companions over a triple vertical member. Weighted Jacobian identities then eliminate all root, collision, and lower-jet branches. On the final branch, where a binary quadratic W_0 is nonzero, the degree-six identity forces the cubic pencil onto the exact nonminimal boundary. Arbitrary quadratic and cubic lower jets are retained throughout.

1 Invariant statement and frozen context

After translating source and target, write a degree-four polynomial map as

$$F = LX + H_2 + H_3 + H_4, \quad L \in \text{GL}_3(\mathbb{C}), \quad (1)$$

where H_i is homogeneous of degree i and $H_4 \neq 0$. The map is *Keller* if $\det JF \in \mathbb{C}^\times$.

For a rank-two H_4 , let

$$h = \gcd(H_{4,1}, H_{4,2}, H_{4,3}), \quad e = \deg h, \quad G = H_4/h.$$

Put

$$K_G = \mathbb{C}(G_i/G_j : G_j \neq 0) \subset M = \mathbb{C}(\mathbb{P}^2),$$

and let E_G be the relative algebraic closure of K_G in M . The curve with function field E_G is rational, so $E_G = \mathbb{C}(p/q)$ for coprime homogeneous forms of a common least degree a . If the

reduced projective image of G has degree δ and $\nu = [E_G : K_G]$, the induced map from the pencil line is represented by a basepoint-free binary triple A of degree

$$b = \delta\nu, \quad e + ab = 4. \quad (2)$$

This defines the invariant tuple in the theorem.

Theorem 1 (Fixed-linear primitive cubic-pencil exclusion). *Let F be as in (1), of exact total degree four, and suppose*

$$(\text{rank } JH_4, e, a, b, \delta, \nu) = (2, 1, 3, 1, 1, 1).$$

If F is Keller, then F is a polynomial automorphism.

Corollary 2. *No degree-four Keller counterexample belongs to frozen row Q2-E1-A3-B1-D1-N1.*

The project's frozen denominator contains fourteen disjoint inclusive leading-form rows. At this release the internal status is

$$3/14 \text{ certified excluded}, \quad 5/14 \text{ provisional}, \quad 6/14 \text{ open}.$$

This is bookkeeping context, not a theorem about the other rows. The universal total-degree floor remains four.

2 Canonical normalization

Proposition 3. *After independent invertible source and target changes,*

$$H_4 = (hp, hq, 0)^T, \quad (3)$$

where h is linear, p, q are coprime nonproportional cubics, and

$$\mathbb{C}(p/q) \text{ is relatively algebraically closed in } \mathbb{C}(\mathbb{P}^2). \quad (4)$$

The target normalization has three intrinsic rank-two charts and does not divide by a frozen coefficient pivot.

Proof. The construction preceding (2) gives the exact polynomial factorization

$$H_4 = hA(p, q).$$

Indeed, the curve with function field E_G is dominated by $\mathbb{P}^2$. Restriction to a general source line makes it dominated by $\mathbb{P}^1$, hence rational. The primitive triples G and $A(p, q)$ define the same projective rational map. The substituted triple is primitive because p, q are coprime and A is basepoint-free. Two primitive triples over the UFD $\mathbb{C}[x, y, z]$ defining the same projective map differ by a unit, which can be absorbed into A .

Here $b = 1$, so

$$A(u, v) = \mathbf{a}u + \mathbf{b}v$$

for linearly independent $\mathbf{a}, \mathbf{b} \in \mathbb{C}^3$. Complete them to a target basis and apply its inverse. This proves (3). Explicitly, the three charts are given by the nonvanishing of

$$\Delta_{01} = a_0b_1 - a_1b_0, \quad \Delta_{02} = a_0b_2 - a_2b_0, \quad \Delta_{12} = a_1b_2 - a_2b_1.$$

Adjoining the unused standard basis vector gives determinant, up to sign, equal to the selected minor. These charts cover the rank-two condition and refer to no first-nonzero coefficient of H_4 . Finally, $a = 3$ is the least pencil degree by the relative-closure construction, giving (4). Arbitrary lower jets are transported bijectively. $\square$

3 Weighted identities and first-integral descent

Put

$$P = hp, \quad Q = hq, \quad D(R) = \text{Jac}(P, Q, R),$$

and introduce

$$\mathcal{J}(\tau) = L + \tau JH_2 + \tau^2 JH_3 + \tau^3 JH_4, \quad E_j = [\tau^j] \det \mathcal{J}(\tau).$$

The Keller condition gives $E_j = 0$ for $j > 0$. If $G_3 = (H_3)_3$ and $G_2 = (H_2)_3$, the zero third row of JH_4 gives

$$E_8 = D(G_3) = 0. \tag{5}$$

After $G_3 = 0$, the third rows of both JH_4 and JH_3 vanish, and

$$E_7 = D(G_2) = 0. \tag{6}$$

Lemma 4 (Homogeneous first-integral descent). *If $0 \neq R \in \mathbb{C}[x, y, z]$ is homogeneous of degree d and $D(R) = 0$, then*

$$\frac{R^4}{P^d} = \mathcal{R}(q/p) \quad \text{for some } \mathcal{R} \in \mathbb{C}(t). \tag{7}$$

Proof. The functions P, Q are algebraically independent: $Q/P = q/p$ is nonconstant on $\mathbb{P}^2$, while P supplies the independent scaling parameter. In characteristic zero, $dP \wedge dQ \wedge dR = 0$ makes R algebraic over $\mathbb{C}(P, Q) = \mathbb{C}(q/p, P)$.

The quotient $\Theta = R^4/P^d$ has degree zero and lies in $M = \mathbb{C}(\mathbb{P}^2)$. With a scaling coordinate s one has $\mathbb{C}(x, y, z) = M(s)$, $P = s^4 P_0$, and $R = s^d R_0$, so P is transcendental over M . Clear denominators in an algebraic relation for Θ over $\mathbb{C}(q/p)(P)$ and collect powers of P . Transcendence makes each coefficient vanish; hence Θ is algebraic over $\mathbb{C}(q/p)$. Relative algebraic closure (4) gives $\Theta \in \mathbb{C}(q/p)$. $\square$

4 The horizontal/unique-vertical split

Let $\mathcal{P} = \langle p, q \rangle$ and restrict to $h = 0$:

$$\rho_h : \mathcal{P} \longrightarrow \mathbb{C}[x, y, z]_3/(h).$$

Rank two means that no pencil member is divisible by h and defines the *horizontal* branch. Rank one gives a unique projective vertical member. Rank zero would make h divide both generators, contrary to coprimality. This split is intrinsic under a change of pencil generators.

4.1 The horizontal branch

Horizontality says that h divides neither p nor $q - \lambda p$ for any $\lambda \in \mathbb{C}$. Factoring the numerator and denominator of $\mathcal{R}$ in (7) therefore gives $v_h(\mathcal{R}(q/p)) = 0$. Since $v_h(P) = 1$,

$$4v_h(R) = d.$$

There is no nonzero homogeneous first integral of degree two or three. Equations (5) and (6) give

$$G_3 = G_2 = 0.$$

The third component of F is linear. After linear changes, $F = (F_1, F_2, z)$; each plane fibre is Keller of degree at most four. The bounded plane theorem used below makes each fibre an automorphism, whence F is injective and therefore an automorphism.

4.2 The vertical branch

Set $h = z$ and choose the unique vertical member as the first generator:

$$p = z^m r, \quad 1 \leq m \leq 3, \quad z \nmid r q. \quad (8)$$

The integer $m = v_z(p)$ is intrinsic.

5 The quadratic-component exit

Lemma 5. *Let $F : \mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ be Keller of degree at most four. If a nonzero target-linear combination of its components has degree at most two, then F is an automorphism.*

Proof. Call the combination f . Its gradient never vanishes because JF is invertible. Write $\nabla f = HX + b$, with H constant symmetric. If $b \in \text{im } H$, the equation $HX = -b$ gives a critical point. Thus $b \notin (\ker H)^\perp$. Choose $v \in \ker H$ with $b^T v \neq 0$. In linear coordinates with v as the third direction,

$$f = g(y_1, y_2) + \beta y_3, \quad \beta \neq 0.$$

The triangular map $T = (y_1, y_2, f)$ and its inverse both have degree at most two. After the target change selecting f , put $G = F \circ T^{-1}$. Then

$$G = (G_1, G_2, x_3), \quad \deg G \leq 8.$$

Each fibre $x_3 = c$ is a plane etale polynomial map of maximum component degree at most eight.

Vistoli states on journal p. 80 that an etale polynomial self-map of $\mathbb{A}^2$ over the algebraically closed characteristic-zero field fixed on p. 79 is an isomorphism when its degree is at most 12 [2, p. 80]. He attributes this theorem to Moh and notes that Moh proves the result through degree 100, whereas 12 is all Vistoli uses. Hence every fibre of G is an automorphism. Equal images under G have equal third coordinates and then equal first two coordinates, so G is injective. The injective-etale theorem quoted by Vistoli, or Ax–Grothendieck together with etaleness, makes G an automorphism. $\square$

Remark 6. This uses Moh’s bounded plane theorem in the exact degree- ≤ 12 form quoted by Vistoli. It does not assume the unresolved plane Jacobian Conjecture.

6 The vertical multiplicity lemma

Lemma 7. *For (8), the cubic kernel of (5) is*

m	$\ker(D : \mathbb{C}[x, y, z]_3 \rightarrow \mathbb{C}[x, y, z]_8)$
1	0
2	0
3	$\langle z^3, q \rangle$.

Proof. Apply Lemma 4 to a nonzero cubic R , and put $\eta = \text{ord}_\infty \mathcal{R}$, with a degree- n polynomial having order $-n$ at infinity. If a prime f occurs in p with multiplicity a , valuation of (7) gives

$$4v_f(R) - 3(a + \mathbf{1}_{f=z}) = a\eta. \quad (9)$$

For $m = 1$, the z -equation gives $\eta \equiv 2 \pmod{4}$. Every prime of the quadratic r has multiplicity $a = 1$ or 2 , but its equation requires $a(3 + \eta) \equiv 0 \pmod{4}$, impossible because $3 + \eta \equiv 1 \pmod{4}$.

For $m = 2$, the z -equation is $4v_z(R) = 9 + 2\eta$, an opposite-parity contradiction. For $m = 3$ it is

$$4v_z(R) = 12 + 3\eta.$$

Since $0 \leq v_z(R) \leq 3$, the possibilities are $(v_z(R), \eta) = (3, 0)$ and $(0, -4)$. The first gives $R \sim z^3$. In the second, $\mathcal{R}$ has no finite pole, since a pole would give a negative valuation along a finite pencil fibre while the left side of (7) is nonnegative there. Thus $\mathcal{R}$ is a degree-four polynomial. A root of multiplicity $n < 4$ would require $4 \mid na$ for every component multiplicity $a \in \{1, 2, 3\}$ of a cubic fibre, impossible in total degree three. Hence $\mathcal{R} = c(t - \lambda)^4$ and unique factorization gives $R \sim q - \lambda z^3$. $\square$

The cases $m = 1, 2$, and $m = 3$ with $G_3 = 0$ are settled by Lemma 5. If $m = 3$ and $G_3 = \alpha z^3 + \beta q \neq 0$, then $\beta = 0$ gives the *vertical companion* $G_3 = z^3$. If $\beta \neq 0$, replacing q by G_3 gives the *nonvertical companion* $G_3 = q$. They cannot merge: the unique vertical member is divisible by z , and q is not.

7 Companion gauges and the root atlas

Write

$$H_3 = (U, V, G_3)^T, \quad H_2 = (A, B, W)^T, \quad \{f, g\} = f_x g_y - f_y g_x.$$

For $G_3 = q$, the complete weight-seven solve modulo legal target shears is

$$U = dz^3, \quad V = zW + fz^3. \quad (10)$$

For $G_3 = z^3$,

$$E_7 = z^3\{q, 4zW - 3U\} = 0,$$

and the complete legal gauge is

$$U = \frac{4}{3}zW + \sigma q, \quad [z^3]V = 0. \quad (11)$$

Target shears remove the free z^3 coefficients of q and V while only renaming unrestricted lower jets. The split $\sigma = 0$ or $\sigma \neq 0$ is exhaustive.

The nonzero binary cubic $q_0 = q|_{z=0}$ has root partition $1 + 1 + 1$, $2 + 1$, or 3 , represented by $xy(x - y)$, x^2y , and x^3 . On the triple-root locus, the full parabolic stabilizer gives the three minimal charts

$$\begin{aligned} q_C &= x^3 + y^2z + \alpha xz^2 + \beta z^3, \\ q_B &= x^3 + xyz + \beta z^3, \\ q_E &= x^3 + yz^2. \end{aligned} \quad (12)$$

The fourth shape is $q \in \mathbb{C}[x, z]_3$ and is exactly the nonminimal boundary. For the vertical companion the term βz^3 is removable; for the nonvertical companion it is retained.

8 The fifteen terminal groups

Let SF, D, and T denote squarefree, double-plus-simple, and triple root types. The following table is disjoint and exhaustive.

ID	Predicate	Exit
H	horizontal restriction rank two	homogeneous valuation, then plane fibres
Z1	vertical $m = 1$	$G_3 = 0$, Lemma 5
Z2	vertical $m = 2$	$G_3 = 0$, Lemma 5
Z3	$m = 3$, $G_3 = 0$	Lemma 5
N1	$G_3 = q$, q_0 SF	nonvertical nontriple obstruction
N2	$G_3 = q$, q_0 D	nonvertical nontriple obstruction
N3	$G_3 = q$, q_0 T	three-chart nonvertical obstruction
V1	$G_3 = z^3$, $\sigma \neq 0$, q_0 SF, $\ell = 0$	zero- ℓ obstruction
V2	same, $\ell \neq 0$	nonzero- ℓ obstruction
V3	$G_3 = z^3$, $\sigma \neq 0$, q_0 D, $\ell = 0$	zero- ℓ obstruction
V4	same, $\ell \neq 0$	nonzero- ℓ , including both collisions
V5	$G_3 = z^3$, $\sigma \neq 0$, q_0 T, $\gamma \neq 0$	direct E_6 contradiction
V6	same, $\gamma = 0$	E_6 forces $\ell = 0$, then E_5, E_4
V7	$G_3 = z^3$, $\sigma = 0$, $W_0 = 0$	five-chart E_6, E_5, E_4 obstruction
V8	$G_3 = z^3$, $\sigma = 0$, $W_0 \neq 0$	E_6 forces the nonminimal boundary

Here $W_0 = W|_{z=0}$. If $\sigma \neq 0$, restriction of E_6 gives

$$-\sigma q_0 \{q_0, W_0\} = 0. \quad (13)$$

For SF or D the degree-two bracket kernel is zero, so $W = z\ell + \omega z^2$. For T, (13) gives $W_0 = \gamma x^2$. Thus the table covers the entire vertical companion.

8.1 Nonvertical terminals N1–N3

With (10), for a noncube q_0 , restrictions of E_6, E_5, E_4 give

$$A_0 = 0, \quad \bar{L}_1 = 0, \quad A = \alpha z^2 \text{ or } B_0 = 0.$$

The two full systems have constant pivot minors -2^{19} and -2^{11} , uniformly for $q_0 = xy(x - y)$ and $q_0 = x^2y$, and force

$$B = z(\ell_{31}x + \ell_{32}y + \beta z), \quad \bar{L}_2 = 0, \quad \bar{L}_3 = (\ell_{31}, \ell_{32}).$$

The first two rows of L are multiples of z , so $\det L = 0$.

On the three charts (12), the full E_6, E_5 systems have constant 14×14 minors

$$-2^{24}3^8, \quad -2^{18}3^6, \quad -2^{20}3^7,$$

and give the same dependent-row conclusion. The moduli α, β are not specialized.

8.2 Vertical terminals V1–V6

For SF or D and $\ell = ux + vy \neq 0$, the complete binary E_5 kernel is $V_0 = \kappa q_0$, $\bar{L}_3 = 0$, except on the two D collision lines, whose larger kernels are retained. Raw coefficients give

$$\begin{array}{l|l} \text{SF} & [x^4yz]E_6 = \sigma u \quad [xy^4z]E_6 = -\sigma v \\ \text{D, noncollision} & [x^4yz]E_6 = \sigma u \quad [x^3y^2z]E_6 = -2\sigma v. \end{array}$$

On the collision kernels $\ell = cx$ and $\ell = cy$, the decisive coefficients are σc and $-2\sigma c$. Hence $\ell \neq 0$ is impossible.

If $\ell = 0$, the complete degree-six solution is

$$V = kq + \frac{z}{\sigma}(A - a_5z^2) - \frac{4}{3\sigma}z^2(\ell_{31}x + \ell_{32}y).$$

Constant pivots $2^53^{11}\sigma^8$ and $-2^43^5\sigma^5$ prove completeness of the E_6, E_5 solves. They give $\ell_{31} = \ell_{32} = 0$, and E_4 gives

$$\ell_{21} = \frac{k}{\sigma}\ell_{11}, \quad \ell_{22} = \frac{k}{\sigma}\ell_{12}.$$

Thus $\det L = 0$.

For T, put $W = \gamma x^2 + z(ux + vy + \omega z)$. If $\gamma \neq 0$, the three charts give

$$\begin{aligned} [x^4yz]E_6 &= 4\gamma\sigma, \\ [x^5z]E_6 &= \sigma(2\gamma - 3v), \quad -\frac{1}{6}[x^3yz^2]E_6 + [xy^2z^3]E_6 = -\frac{\sigma}{3}(\gamma + v), \\ \frac{2}{3}[x^4z^2]E_6 + [xyz^4]E_6 &= \frac{10}{3}\gamma\sigma. \end{aligned}$$

Each chart is impossible. If $\gamma = 0$, every chart first gives $[x^5z]E_6 = -3\sigma v$, and a second chartwise linear combination gives a nonzero constant times σu . Hence $u = v = 0$, after which the complete E_6, E_5, E_4 solution above makes $\det L = 0$.

8.3 The zero-parameter, zero- W_0 terminal V7

Write $W = z(ux + vy + \omega z)$. The SF, D, and three minimal T charts form a complete atlas. A constant 5×5 subsystem of E_6 gives on all five charts

$$\begin{aligned} A_{20} &= \frac{2}{9}u^2, & A_{11} &= \frac{4}{9}uv, & A_{02} &= \frac{2}{9}v^2, \\ \ell_{31} &= \frac{9A_{10} - 4u\omega}{12}, & \ell_{32} &= \frac{9A_{01} - 4v\omega}{12}. \end{aligned}$$

The nontriple charts and q_C force $u = v = 0$ through cubic monomials of E_5 . The q_B, q_E charts either do the same or leave one explicit $u \neq 0$ compatibility family. On $u = v = 0$, E_5, E_4 force

$$\ell_{11} = \ell_{12} = \ell_{31} = \ell_{32} = 0,$$

so $\det L = 0$. On either exceptional $u \neq 0$ family, two or three raw E_4 coefficients require incompatible values of a coefficient of V .

9 The final $W_0 \neq 0$ elimination

Assume $\sigma = 0$, so $U = \frac{4}{3}zW$. Row multilinearity gives

$$3E_6 = z\Phi, \quad \Phi = 4W\{q, W\} + 9z^2\{A, q\} + 12z^3\{q, L_3\}. \quad (14)$$

The variables B, V, L_1, L_2 cancel or do not occur. The lowest z -coefficient gives

$$\{q_0, W_0\} = 0.$$

For binary forms of degrees three and two, dehomogenization gives $2F'G - 3FG' = 0$. Thus F^2/G^3 is constant, and unique factorization gives

$$q_0 = \kappa L^3, \quad W_0 = \gamma L^2, \quad \kappa\gamma \neq 0. \quad (15)$$

After a binary change take $L = x$ and retain every coefficient:

$$\begin{aligned} q &= \kappa x^3 + z(\alpha x^2 + \beta xy + \chi y^2) + z^2(\delta x + \epsilon y) + \phi z^3, \\ W &= \gamma x^2 + z(ux + vy) + \omega z^2, \\ A &= a_{20}x^2 + a_{11}xy + a_{02}y^2 + z(a_{10}x + a_{01}y) + a_{00}z^2. \end{aligned}$$

Six raw coefficients of Φ suffice. The first four give

$$\chi = 0, \quad r = 2\beta\gamma - 3\kappa v = 0, \quad f = 27\kappa a_{02} + 2\beta\gamma v - 6\kappa v^2 = 0, \quad h = 9a_{02}\beta - 2\beta v^2 = 0.$$

Put $g = 9a_{02} - v^2$. The division-free combinations

$$f - vr = 3\kappa g, \quad h - \beta g = -\beta v^2$$

give $g = 0$ and $\beta v^2 = 0$. Multiplying $r = 0$ by v^2 gives $-3\kappa v^3 = 0$, hence

$$v = \beta = a_{02} = 0.$$

The remaining two coefficients reduce to

$$27\kappa a_{11} + 8\gamma^2\epsilon = 0, \quad 9a_{11}\epsilon = 0.$$

Multiplying the first by ϵ and subtracting three κ times the second gives $8\gamma^2\epsilon^2 = 0$. Therefore $\epsilon = a_{11} = 0$, and

$$q = \kappa x^3 + \alpha x^2 z + \delta x z^2 + \phi z^3 \in \text{Sym}^3\langle x, z \rangle.$$

For a cubic pencil containing z^3 , this is exactly the nonminimal boundary $q \in \text{Sym}^3\langle z, L \rangle$. Its canonical pencil degree is $a = 1$, not $a = 3$, so it reclassifies into a separate frozen row. No point of the minimal row survives. This closes V8 and proves Theorem 1.

10 Frozen pivots, retained calculations, and verification boundary

The frozen coefficient partition orders the fifteen quartic monomials component by component and takes the first nonzero coefficient among $c_0, \dots, c_{44}$. Proposition 3 and the proof above are

global on the invariant row and divide by none of these coefficients. Every nonempty pivot C00–C29 therefore enters the theorem directly. If the first pivot were C30 or later, the first two target components of H_4 would vanish, forcing $\text{rank } JH_4 \leq 1$. Hence C30–C44 are empty.

The post-freeze bridge checker expands the proof into 48 disjoint route atoms mapping uniquely to the fifteen terminal groups. It checks all 45 pivots and includes leading-form witnesses for C00–C29; those witnesses are not asserted to be Keller completions.

Detailed derivations and separate hostile reconstructions are retained in the repository files `WORKING_HORIZONTAL_FIXED_LINEAR_CUBIC_PENCIL.md`, `vertical_locus/WORKING_VERTICAL_FIXED_LINEAR_CUBIC_PENCIL.md`, the branch lemmas under `vertical_locus/`, `vertical_locus/a0_w0_nonzero_attack/NOTE.md`, and `../WORKING_QUADRATIC_COMPONENT_EXIT.md`. The final bridge report is `../taxonomy_freeze/audit_bridge_q2_e1_a3_b1_d1_n1_v1/REPORT.md`. The aggregate entry point is `verify_all_strict.sh`.

These programs check encoded algebra, rank certificates, collision strata, coverage, and deliberate fault injections. Their “hostile audit” label denotes an internal adversarial workflow, not independent human peer review. The literature theorems used in Lemma 5 remain black boxes.

11 Prior art, attribution, and disclosure

The external bounded-degree input is Moh’s plane theorem in the degree- ≤ 12 form quoted by Vistoli. Moh’s cited paper is the 1983 paper on configurations of roots [1]. We do not attribute the present three-dimensional row decomposition to Moh, and we do not confuse the plane theorem with the separate unpublished Moh–Sathaye computer calculation for three-dimensional degree three discussed by Vistoli.

No claim is made that this row theorem is absent from all prior literature. The project’s source-specific search and internal provenance record are not a worldwide priority determination. A fuller account of the AI roles and limitations is in `AI_DISCLOSURE.md`. No external person was contacted by an AI system on behalf of this project.

References

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